

SAGAR R

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PROFILE SUMMARY

Final-year Artificial Intelligence and Data Science Engineering student (CGPA: 9.0/10) with hands-on experience building cross-platform mobile apps, web applications, and low-code enterprise tools. Proficient in Python and SQL, with working knowledge of data visualization platforms (Power BI, Tableau) and cloud services (Microsoft Azure). Seeking a Machine Learning Engineer, Data Scientist, Data Analyst, or AI Research Intern role to apply strong programming, data-handling, and problem-solving skills to real-world challenges.

EDUCATION

East West Institute of Technology, Bengaluru 2023 - 2027
B.E. in Artificial Intelligence and Data Science Engineering
CGPA: 9.0 / 10.0 (through 3rd year)

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C, Dart
Web Technologies: HTML, CSS
Databases: SQL, MongoDB
Tools & Platforms: Git/GitHub, Google Colab, Android Studio, Docker, OutSystems
Cloud & Data Visualization: Microsoft Azure, Power BI, Tableau

PROJECTS

- Smart City Traffic & Emergency Routing** *Python, Dijkstra/A*, OSMnx, Streamlit*
- Built a route-optimization system that computes the fastest path for emergency vehicles (ambulance, fire, police) across a city road network, accounting for live traffic congestion.
 - Modeled the road network as a weighted graph and implemented Dijkstra and A* shortest-path algorithms to recompute optimal routes in real time as traffic conditions changed.
 - Simulated traffic-signal prioritization along the emergency vehicle's route to reduce intersection delay, and built a dashboard to visualize live position, congestion, and recommended path.
- Micro-Grid Energy Arbitrage** *Python, Pandas, Linear Programming, Time-Series Forecasting*
- Built a decision-support model for a solar-plus-battery micro-grid that schedules when to charge, discharge, or sell electricity to take advantage of fluctuating grid prices.
 - Developed a time-series forecasting module to predict hourly electricity price and demand, used as input to the charge/discharge scheduling model.
 - Formulated a linear-programming model to maximize arbitrage profit under battery capacity and charge/discharge-rate constraints, and benchmarked results against a naive always-on baseline.
- On-Device Dietary Volume Estimator** *Python, TensorFlow Lite, Computer Vision, Android*
- Built a mobile computer-vision app that estimates the volume of food on a plate from a single smartphone photo, to support dietary and calorie tracking.
 - Trained a lightweight food-segmentation model and combined it with reference-object scaling to estimate portion volume without requiring a depth sensor.
 - Converted the model to TensorFlow Lite and optimized it for fast on-device inference, so estimation works fully offline without sending photos to the cloud.

CERTIFICATIONS & ACHIEVEMENTS

- Web Programming — Ethnotech Academic Solutions (Sep 2023 – Jan 2024)
- Web Technologies — Ethnotech Academic Solutions (Mar 2024 – Jun 2024)
- Data Science Using Python — Ethnotech Academic Solutions (Aug 2024 – Dec 2024)
- Advanced Java Programming — Ethnotech Academic Solutions (Feb 2025 – May 2025)
- Python for Data Science — NPTEL Elite Certification, IIT Madras (Top 2%, 85%, Jul – Aug 2025)